Curriculum and Reading Roadmap

Quantum Information, Holographic Geometry, Tensor Networks, and Galactic Kinematics

Theoretical and Computational Study Base for the ER $=$ EPR **Galactic Dynamics Model**

Abstract

This curriculum establishes a structured, ten-month academic study plan covering the mathematical, physical, and computational prerequisites necessary to analyze and advance emergent spacetime models of galactic dynamics. The plan links continuous quantum information theory (operator algebras, modular theory, and entropy inequalities), holographic duality (AdS geometry and Ryu–Takayanagi extremal surfaces), discrete tensor network realizations (MERA, HaPPY code, and graph min-cuts), non-local geometric defects (ER $=$ EPR and the Fractal Entanglement Quantum Gravity framework), and observational validation against the SPARC galactic database.

Phase 1: Quantum Information Theory and Many-Body Entanglement

- **Core Concepts:**
  - Trace-class operators $\mathcal{B}_1(\mathcal{H})$ on Hilbert spaces and density matrices ρ.
  - Reduced density operators $\rho_A = \mathrm{Tr}_B(\rho_{AB})$ and von Neumann entanglement entropy $S(A) = -\mathrm{Tr}(\rho_A \ln \rho_A)$.
  - Srednicki's ground-state area law $S(A) \propto \mathrm{Area}(\partial A)$ in gapped Hamiltonians and logarithmic corrections in critical 1D conformal systems.
  - Quantum mutual information $I(A:B) = S(A) + S(B) - S(AB)$, relative entropy $S(\rho\|\sigma)$, and strong subadditivity (SSA).
  - Tomita–Takesaki modular theory, the modular Hamiltonian $K = -\ln \Delta$, and the First Law of Entanglement Entropy $\delta S = \delta\langle K\rangle$.
- **Textbooks:**
  - M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (Chapters 8, 11, and 12).
  - J. Preskill, *Lecture Notes for Physics 229: Quantum Information and Computation*, California Institute of Technology (Chapters 2 and 10).
- **Key Papers:**
  - M. Srednicki, *Entropy and Area*, Phys. Rev. Lett. **71**, 666 (1993).
  - P. Calabrese and J. Cardy, *Entanglement entropy and quantum field theory*, J. Stat. Mech. **0406**, P06002 (2004).

Phase 2: Differential Geometry, Holography, and Ryu–Takayanagi (RT)

- **Core Concepts:**
  - Riemannian geometry of the Poincaré disk $\mathbb{H}^2$ and constant negative sectional curvature.
  - Geodesic boundary-orthogonality in hyperbolic manifolds.
  - Calculus of variations for codimension-2 extremal surfaces and the vanishing mean curvature condition $\mathbf{H} = 0$.

  - The Ryu–Takayanagi prescription: $S(A) = \text{Area}(\gamma_A)/(4G\hbar)$.
  - Entanglement wedge reconstruction and bulk-to-boundary operator mapping.

- **Textbooks:**
  - M. Rangamani and T. Takayanagi, *Holographic Entanglement Entropy*, Lecture Notes in Physics, Vol. 931, Springer (2017).
  - S. M. Carroll, *Spacetime and Geometry: An Introduction to General Relativity*, Cambridge University Press (Chapters 2, 3, and 8).

- **Key Papers:**
  - S. Ryu and T. Takayanagi, *Holographic Derivation of Entanglement Entropy from AdS/CFT*, Phys. Rev. Lett. **96**, 181602 (2006).
  - D. L. Jafferis, A. Lewkowycz, J. Maldacena, and S. J. Suh, *Relative entropy equals bulk relative entropy*, JHEP **06**, 004 (2016).

Phase 3: Discrete Tensor Networks and Graph Algorithms

- **Core Concepts:**
  - Matrix Product States (MPS) and Multi-scale Entanglement Renormalization Ansatz (MERA).
  - Hyperbolic uniform tilings $\{p, q\}$ and exact holographic quantum error-correcting codes (HaPPY pentagon code).
  - Ford–Fulkerson max-flow/min-cut duality: translating boundary entropy calculation into network flow optimization.
  - Push-Relabel algorithms on capacitated graphs.
  - Bond dimension bounds: edge capacities defined by $C_e = \ln \chi_e$.

- **Textbooks and Reviews:**
  - R. Orús, *A Practical Introduction to Tensor Networks: Matrix Product States and Projected Entangled Pair States*, Ann. Phys. **349**, 117–158 (2014).
  - T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, MIT Press (Chapter 26: Maximum Flow).

- **Key Papers:**
  - G. Vidal, *Class of Quantum Many-Body States That Can Be Efficiently Simulated*, Phys. Rev. Lett. **101**, 110501 (2008).
  - F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, *Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence*, JHEP **06**, 149 (2015).

Phase 4: Emergent Gravity, ER $=$ EPR, and Fractal Entanglement

- **Core Concepts:**
  - The Maldacena–Susskind ER $=$ EPR conjecture: identifying non-traversable Einstein–Rosen bridges with EPR entangled states without gauge couplings.
  - Jacobson's thermodynamic field equations: deriving $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ as an equation of state via $\delta Q = T dS$.
  - Verlinde's volume-law entropic displacement and the characteristic acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2.
  - Giné's Fractal Entanglement Quantum Gravity (FEQG) relational distance: $d_{ij} = -l_0 \ln(I_{ij}/I_0)$.
  - The informational stress-energy tensor: $E_{\mu\nu} = \nabla_\mu \nabla_\nu S_{\text{ent}} - g_{\mu\nu} \Box S_{\text{ent}}$ and modified weak-field potentials $V(r) \sim 1/r^{1+\epsilon}$.

- **Key Papers:**
  - T. Jacobson, *Thermodynamics of Spacetime: The Einstein Equation of State*, Phys. Rev. Lett. **75**, 1260 (1995).
  - J. Maldacena and L. Susskind, *Cool horizons for entangled black holes*, Fortschr. Phys. **61**, 781–811 (2013).
  - E. P. Verlinde, *Emergent Gravity and the Dark Universe*, SciPost Phys. **2**, 016 (2017).
  - J. Giné, *Quantum Gravity from Fractal Entanglement Geometry*, arXiv:2607.26035 [gr-qc] (2026).

Phase 5: Observational Galactic Kinematics and Statistical Testing

- **Core Concepts:**
  - Semiclassical circular orbital dynamics: $v_c^2(r) = r\,\partial\Phi_{\text{eff}}/\partial r$.
  - Baryonic velocity decomposition: $v_{\text{bary}}^2 = |V_{\text{gas}}|V_{\text{gas}} + \Upsilon_{\text{disk}}|V_{\text{disk}}|V_{\text{disk}} + \Upsilon_{\text{bul}}|V_{\text{bul}}|V_{\text{bul}}$.
  - Stellar mass-to-light ratio modeling $\Upsilon_\star$ in near-infrared ($3.6\,\mu$m).
  - Baryonic Tully–Fisher Relation (BTFR) and the Radial Acceleration Relation (RAR).
  - Goodness-of-fit benchmarking: reduced chi-square χ^2_ν, residual distributions $\Delta v = v_{\text{obs}} - v_{\text{pred}}$, and comparison against standard NFW and MOND profiles.
- **Textbooks:**
  - J. Binney and S. Tremaine, *Galactic Dynamics*, 2nd Edition, Princeton University Press (2008).
  - P. C. Gregory, *Bayesian Logical Data Analysis for the Physical Sciences*, Cambridge University Press (2005).
- **Key Papers and Catalogs:**
  - F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, Astron. J. **152**, 157 (2016).
  - S. S. McGaugh, F. Lelli, and J. M. Schombert, *Radial Acceleration Relation in Rotationally Supported Galaxies*, Phys. Rev. Lett. **117**, 201101 (2016).

Structured Ten-Month Execution Schedule

Period	Phase Focus	Core Mathematical Objectives	Computational Milestones
Months 1–2	Phase 1: Quantum Information	Trace-class states, strong subadditivity, modular Hamiltonians	Implement 1D coupled harmonic lattice; compute discrete $S(A)$.
Months 3–4	Phase 2: Holographic Geometry	Geodesics on $\mathbb{H}^2$, area functionals, RT minimal cuts	Verify RT area curves analytically and numerically on AdS_3/CFT_2.
Months 5–6	Phase 3: Tensor Networks	Network flow algorithms, MERA disentanglers, HaPPY code	Build hyperbolic graph generator; implement Push-Relabel min-cut solver.
Months 7–8	Phase 4: Emergent Gravity	Giné's master equation, $E_{\mu\nu}$ informational stress tensor	Inject non-local chordal defects; compute effective modified potential $V(r)$.
Months 9–10	Phase 5: Empirical Benchmarks	Kinematic baryonic decomposition, parameter optimization ($\Upsilon_\star, a_0$)	Run full SPARC batch ($N = 171$); generate scaling curves and χ^2_ν distributions.

Table 1: Progression timeline aligning theoretical literature study with concrete numerical deliverables.